



KNOX GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT

2004
TRIAL HSC EXAMINATION

Mathematics

Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks (120)

- Attempt Questions 1-8
- All questions are of equal value
- Use a **SEPARATE** Writing Booklet for each question
- Write your **Student Number** and **Teacher's Initials** on the front cover of each writing booklet

NAME: _____

TEACHER: _____

Total marks (120)

Attempt questions 1 – 8

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks)	Use a SEPARATE writing booklet.	Marks
(a) Evaluate $\int_0^{1.5} \frac{2}{\sqrt{9-x^2}} dx$.		2
(b) Find $\int \frac{1}{\sqrt{x^2-4x+5}} dx$, with the aid of the Table of Standard Integrals.		2
(c) Find $\int \sin^2 x \cos^3 x dx$.		3
(d) Using the substitution $x = 3\sec\theta$, evaluate $\int_3^6 \frac{1}{x^2\sqrt{x^2-9}} dx$.		4
(e) (i) Find constants A, B, C such that $\frac{x^2+2}{x^2-x-2} \equiv A + \frac{Bx+C}{x^2-x-2}$.		1
(ii) Hence find $\int \frac{x^2+2}{x^2-x-2} dx$.		3

Question 2 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Simplify $\frac{1-i^3}{1-i}$. 2

(b) Let $z = \frac{8-i}{2+i}$.

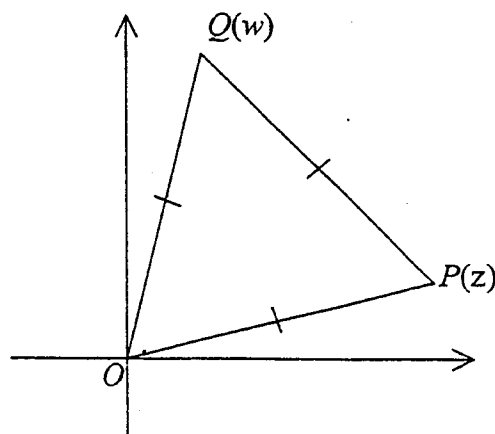
(i) Express z in the form $a+bi$ where a and b are real numbers. 2

(ii) Hence, or otherwise, find $|z|$ and $\arg z$ (to 3 significant figures in the domain $-\pi < \theta \leq \pi$). 3

(c) Sketch the region in the complex number plane where the inequalities $|z+1-2i| \leq 2$ and $\operatorname{Re}(z) \leq 0$ hold simultaneously. 2

(d) Factorise $x^4 + 7x^2 - 18$ into the product of linear factors over the complex field. 2

(e)



In the Argand diagram, OPQ is an equilateral triangle. P represents the complex number z and Q represents the complex number w .

(i) Explain why $w = z \operatorname{cis} \frac{\pi}{3}$. 2

(ii) Show that $w^3 + z^3 = 0$. 2

- (a) Let $f(x) = 2(x-1)(x-3)$.

Draw separate sketches of the following functions (at least one-third of a page), showing clearly the important features, including any intercepts on the axes, turning points, asymptotes, etc.

- | | | |
|-------|----------------------|---|
| (i) | $y = f(x)$ | 1 |
| (ii) | $y = \frac{1}{f(x)}$ | 2 |
| (iii) | $y = 2 - f(x)$ | 2 |
| (iv) | $y = \sqrt{f(x)}$ | 2 |
| (v) | $y = \log_e f(x)$ | 2 |

- (b) Let $I_n = \int_0^1 x^n e^{-x} dx$.

- | | | |
|-------|---|---|
| (i) | Evaluate I_0 . | 1 |
| (ii) | Prove that $I_n = n I_{n-1} - \frac{1}{e}$ for $n \geq 1$. | 3 |
| (iii) | Hence evaluate $\int_0^1 x^3 e^{-x} dx$. | 2 |

Question 4 (15 marks) Use a SEPARATE writing booklet.

Marks

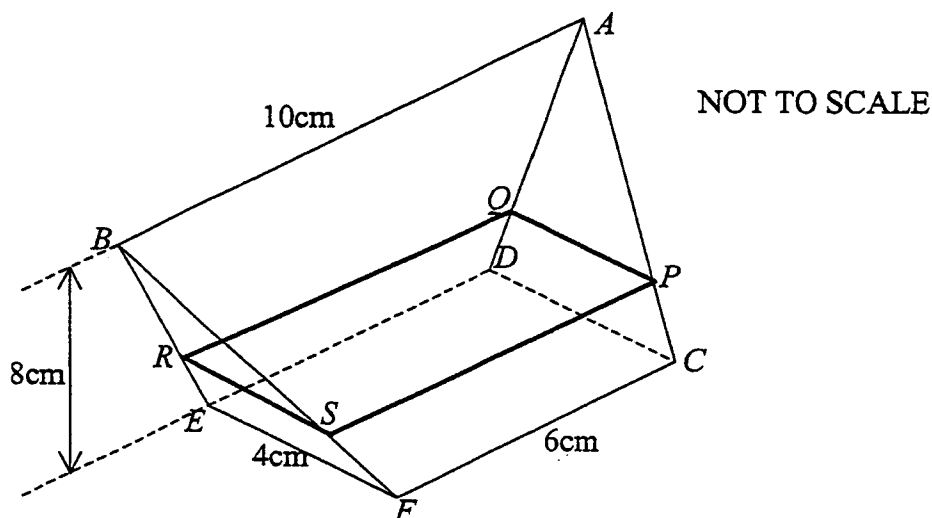
- (a) Find $\sqrt{9-12i}$. 3
- (b) $(2+i)$ is a zero of the polynomial $P(z) = z^3 - z^2 + az + b$, where a and b are real numbers. 4
- Find the other two zeros, and the values of a and b .
- (c) α, β, γ are the roots of the equation $x^3 - 6x^2 + 12x - 35 = 0$.
- (i) Form a cubic equation whose roots are $\alpha - 2, \beta - 2, \gamma - 2$. 2
- (ii) Hence, or otherwise, solve the equation $x^3 - 6x^2 + 12x - 35 = 0$ over the complex field. 2
- (d) The roots of the equation $z^2 + 5z - 2i = 0$ are α and β . Without solving this equation, form the cubic equation whose roots are α, β and $(\alpha + \beta)$. 4

(a) Consider the hyperbola $\frac{x^2}{4} - \frac{y^2}{16} = 1$.

(i) Find its eccentricity. 1

(ii) State the equations of the asymptotes. 1

(b)



The diagram shows a wedge with the edge AB parallel to the horizontal rectangular base $CDEF$, and the plane $ABED$ is vertical. AB is 8 cm vertically above DE . $PQRS$ is a rectangular cross-section h cm above the base.

(i) Show that the area of the cross-section $PQRS$ is $\left(6 + \frac{h}{2}\right)\left(4 - \frac{h}{2}\right) \text{ cm}^2$. 2

(ii) Hence find the volume of the wedge. 2

(c) Consider the function $y = \frac{x^2 - 3x}{x + 1}$.

(i) Find the equations of the two asymptotes. 2

(ii) Find the coordinates of the stationary points and determine their nature. 5

(iii) Sketch the graph of the function. 1

(iv) For what values of k does the equation $\frac{x^2 - 3x}{x + 1} = k$ have two real roots? 1

Question 6 (15 marks) Use a SEPARATE writing booklet.

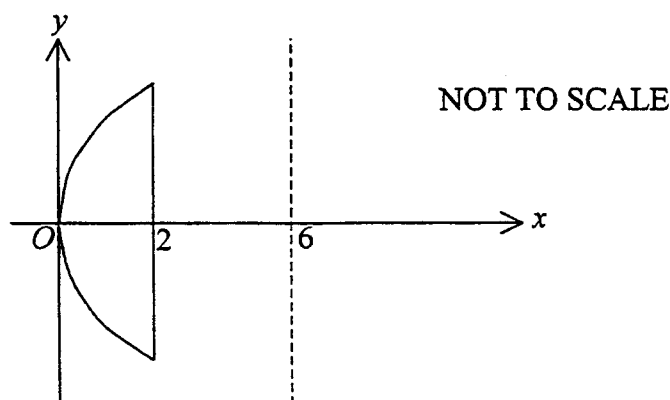
Marks

(a) (i) Show that $f(x) = x\sqrt{4-x^2}$ is an odd function. 1

(ii) Hence, without finding any primitives, evaluate 2

$$\int_{-2}^2 (x\sqrt{4-x^2} - \sqrt{4-x^2}) dx, \text{ giving reasons.}$$

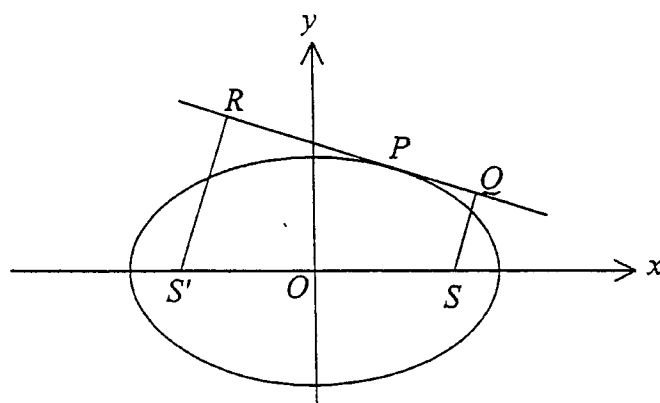
(b)



The region bounded by the parabola $y^2 = 4x$ and the line $x = 2$ is rotated about the line $x = 6$. 5

Using the method of cylindrical shells, find the volume of the solid formed.

(c)



(i) Prove that the equation of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $P(a \cos \theta, b \sin \theta)$ is $(b \cos \theta)x + (a \sin \theta)y - ab = 0$. 3

(ii) Q and R are the feet of the perpendiculars to the tangent from the foci S and S' respectively. 4

Prove that $SQ \times S'R = b^2$.

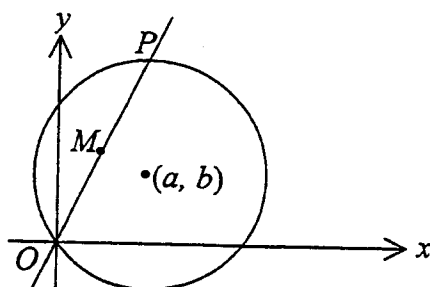
Question 7 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) Find the general solution of the inequality $\cos \theta \geq \frac{1}{2}$.

2

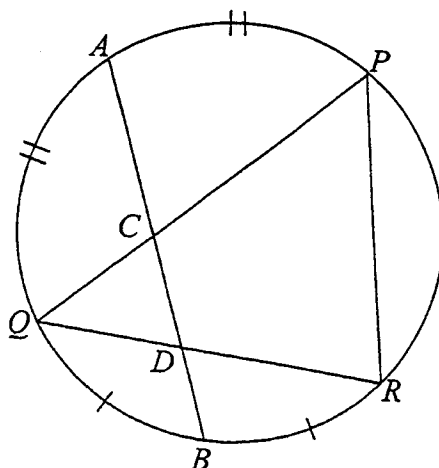
(b)



The diagram shows the graph of the circle $(x-a)^2 + (y-b)^2 = a^2 + b^2$, which passes through the origin O . The line $y = mx$ cuts the circle at O and P .

- (i) Show that the x coordinate of P is $\frac{2(a+bm)}{1+m^2}$. 2
- (ii) Hence write down the coordinates of M , the midpoint of OP . 2
- (iii) Hence show that the locus of M , as the gradient of OP varies, is a circle, and state its centre and radius. 4

(c)



A circle is drawn through the vertices of the triangle PQR . A is the midpoint of the arc PQ and B is the midpoint of the arc QR . The chord AB intersects PQ at C and QR at D .

Copy or trace the diagram into your Writing Booklet.

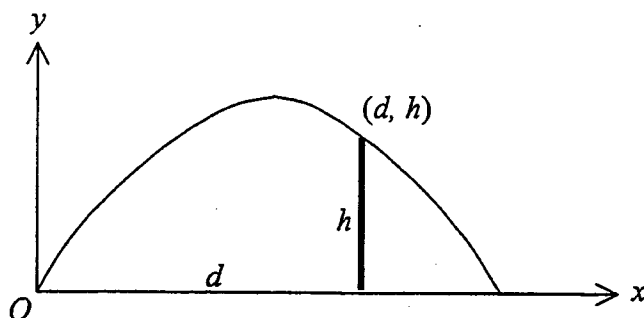
- (i) Explain why $\angle QPB = \angle BPR$.

1

- (ii) Prove that $QC = QD$.

4

(a)



A stone is projected from a point on the ground, and it just clears a fence d metres away. The height of the fence is h metres. The angle of projection to the horizontal is θ and the speed of projection is V m/s.

The displacement equations, measured from the point of projection, are:

$$x = V \cos \theta t \quad \text{and} \quad y = V \sin \theta t - \frac{1}{2} g t^2.$$

(i) Show that $V^2 = \frac{gd^2 \sec^2 \theta}{2(d \tan \theta - h)}$. 2

(ii) Show that the maximum height reached is $\frac{d^2 \tan^2 \theta}{4(d \tan \theta - h)}$. 3

(iii) Show that the stone will just clear the fence at its highest point if $\tan \theta = \frac{2h}{d}$. 3

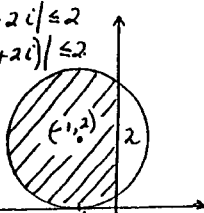
(b) (i) Prove by mathematical induction that $(\sqrt{3} - 1)^n = p_n + q_n \sqrt{3}$, where n is a positive integer and p_n and q_n are unique integers. 5

(ii) Hence show that $p_n^2 - 3q_n^2 = (-2)^n$. 2

End of paper

Suggested Solution (s)	Comments
<u>QUESTION 1</u>	
$(a) \int_0^{1.5} \frac{2}{\sqrt{9-x^2}} = 2 \left[\sin^{-1} \frac{x}{3} \right]_0^{1.5} \quad \checkmark$ $= 2 \left[\sin^{-1} \frac{1}{2} - \sin^{-1} 0 \right]$ $= 2 \left[\frac{\pi}{6} - 0 \right]$ $= \frac{\pi}{3} \quad \checkmark \quad (2)$	
$(b) \int \frac{1}{\sqrt{x^2-4x+5}} dx = \int \frac{1}{\sqrt{(x-2)^2+1}} dx \quad \checkmark$ $= \log_e(x-2 + \sqrt{(x-2)^2+1}) + C$ $\text{or } \log_e(x-2 + \sqrt{x^2-4x+5}) + C \quad (2)$	Ignore 'c' in marking.
$(c) \int \sin^2 x \cos^3 x dx$ $= \int \sin^2 x (1 - \sin^2 x) \cos x dx$ $= \int u^2 (1 - u^2) du \quad \checkmark$ $= \int (u^2 - u^4) du$ $= \frac{1}{3} u^3 - \frac{1}{5} u^5 + C \quad \checkmark$ $= \frac{1}{3} \sin^3 x - \frac{1}{5} \sin^5 x + C \quad \checkmark \quad (3)$	$u = \sin x$ $du = \cos x dx$ or equivalent without substitution
$(d) \int_3^6 \frac{1}{x^2 \sqrt{x^2-9}} dx$ $= \int_0^{\frac{\pi}{3}} \frac{3 \sec \theta \tan \theta d\theta}{9 \sec^2 \theta \sqrt{9 \sec^2 \theta - 9}} \quad \checkmark$ $= \int_0^{\frac{\pi}{3}} \frac{3 \sec \theta \tan \theta d\theta}{9 \sec^2 \theta \times 3 \tan \theta}$ $= \frac{1}{9} \int_0^{\frac{\pi}{3}} \cos \theta d\theta \quad \checkmark$ $= \frac{1}{9} [\sin \theta]_0^{\frac{\pi}{3}} = \frac{\sqrt{3}}{18} \quad \checkmark \quad (4)$	$x = 3 \sec \theta$ $dx = 3 \sec \theta \tan \theta d\theta$ $x = 3, \theta = 0$ $x = 6, \theta = \frac{\pi}{3}$

Suggested Solution (s)	Comments
<u>QUESTION 1 (cont.)</u>	
$(e) (i) \frac{1}{x^2-x-2} = \frac{1}{x^2+0x+2}$ $\frac{x^2-x-2}{x^2-x-2} = \frac{x^2+0x+2}{x^2-x-2}$ $\frac{x^2+2}{x^2-x-2} = 1 + \frac{x+4}{x^2-x-2}$ $A=1, B=1, C=4 \quad (1)$	<u>OR</u> $\frac{x^2+2}{x^2-x-2} = A + \frac{Bx+C}{x^2-x-2}$ $x^2+2 = A(x^2-x-2) + Bx+C$ - equate coefficients
$(ii) \frac{x+4}{(x-2)(x+1)} = \frac{A}{x-2} + \frac{B}{x+1}$ $x+4 = A(x+1) + B(x-2)$ $\text{Sub. } x=-1: 3 = -3B$ $\therefore B = -1$ $\text{Sub. } x=2: 6 = 3A$ $A = 2$ $\therefore \int \frac{x^2+2}{x^2-x-2} = \int \left(1 + \frac{2}{x-2} - \frac{1}{x+1} \right) dx$ $= x + 2 \ln(x-2) - \ln(x+1) + C \quad (3)$	1 mark - correct A, B. 2 marks for 3 correct primitives

Suggested Solution (s)	Comments
<u>QUESTION 2</u> a) $\frac{1-i}{1-i} = \frac{1+i}{1-i} \times \frac{1+i}{1+i}$ OR $\frac{(1-i)(1+i+i^2)}{1-i}$ ✓ $= \frac{1+2i+i^2}{1-i^2} = \frac{1+i-1}{1-(-1)} = \frac{i}{2}$ ✓ (b) $z = \frac{8-i}{2+i}$ (i) $z = \frac{8-i}{2+i} \times \frac{2-i}{2-i}$ $= \frac{16-10i+i^2}{4-i^2} = \frac{15-10i}{5} = 3-2i$ ✓ (ii) $z = \sqrt{3^2+(-2)^2} = \sqrt{13}$ ✓ $\tan \theta = -\frac{2}{3}$ ✓ $\theta = -0.588$ ✓ (c) $z+1-2i \leq 2$ $z-(-1+2i) \leq 2$  $\text{Re}(z) \leq 0$ $x \leq 0$ 1 - circle correct 1 - shading correct.	(2) (2) (3) (2)

Suggested Solution (s)	Comments
<u>QUESTION 2 (cont)</u> (d) $x^4 + 7x^2 - 18$ $= (x^2+9)(x^2-2)$ $= (x^2-9i^2)(x^2-2)$ $= (x-3i)(x+3i)(x-\sqrt{2})(x+\sqrt{2})$ (2) (e) (i) Since $OP = OQ$ and $\angle POQ = \frac{\pi}{3}$, multiplying z by $\text{cis } \frac{\pi}{3}$ doesn't change the modulus, as $\text{cis } \frac{\pi}{3} = 1$, but adds $\frac{\pi}{3}$ to the argument. (2) (ii) $w = z \text{cis } \frac{\pi}{3}$ $w^3 = (z \text{cis } \frac{\pi}{3})^3$ $= z^3 \text{cis } \pi$ $= z^3 (\cos \pi + i \sin \pi)$ $= z^3 (-1 + 0i)$ $= -z^3$ $\therefore w^3 + z^3 = 0$ ✓ (2)	1 - for factors of x^2+9 1 - for factors of x^2-2 1 - modulus reason 1 - argument reason

Suggested Solution (s)	Comments
<u>QUESTION 3.</u> a) $f(x) = 2(x-1)(x-3)$ (i) $y = f(x)$ ① (ii) $y = \frac{1}{f(x)}$ ② (iii) $y = 2 - f(x)$ ② (iv) $y = \sqrt{f(x)}$ ②	

Suggested Solution (s)	Comments
<u>QUESTION 3 (cont.)</u> a)(v) $y = \log_e f(x)$ ② (b) $I_n = \int_0^1 x^n e^{-x} dx$ (i) $I_0 = \int_0^1 e^{-x} dx$ $= [-e^{-x}]_0^1$ $= (-e^{-1}) - (-e^0)$ $= 1 - \frac{1}{e}$ ① (ii) $I_n = \int_0^1 x^n \frac{d}{dx}(e^{-x}) dx$ $= [-x^n e^{-x}]_0^1 - \int_0^1 (-e^{-x}) n x^{n-1} dx \checkmark$ $= [-e^{-1} - 0] + n \int_0^1 x^{n-1} e^{-x} dx \checkmark$ $= -\frac{1}{e} + n I_{n-1}$ ③ (iii) $\int_0^1 x^3 e^{-x} dx = I_3$ $= 3 I_2 - \frac{1}{e}$ $= 3 [2 I_1 - \frac{1}{e}] - \frac{1}{e} \checkmark$ $= 6 I_1 - \frac{4}{e}$ $= 6 [1 \times I_0 - \frac{1}{e}] - \frac{4}{e}$ $= 6 I_0 - \frac{10}{e}$ $= 6 (1 - \frac{1}{e}) - \frac{10}{e}$ $= 6 - \frac{16}{e} \checkmark$ ②	

Suggested Solution (s)	Comments
<u>QUESTION 4.</u> a) Let $9-12i = x+yi$ (x, y real) $9-12i = x^2-y^2+2xyi$ $x^2-y^2=9, \quad 2xy=-12 \quad \checkmark$ $x^2-\frac{36}{x^2}=9 \quad y=-\frac{6}{x}$ $x^4-9x^2-36=0$ $(x^2-12)(x^2+3)=0$ $x=\pm 2\sqrt{3} \quad (x \text{ real}) \quad \checkmark$ If $x=2\sqrt{3}, y=-\frac{6}{2\sqrt{3}}=-\sqrt{3}$ If $x=-2\sqrt{3}, y=-\frac{6}{-2\sqrt{3}}=\sqrt{3}$ $\therefore \sqrt{9-12i} = \pm(2\sqrt{3}-\sqrt{3}i) \quad \checkmark \quad (3)$ b) $P(z) = z^3 - z^2 + az + b$. $(2+i)$ is a root $\therefore (2-i)$ is a root $\checkmark$ By sum of roots: $(2+i)+(2-i)+\alpha = 1$ $4+\alpha = 1$ $\alpha = -3 \quad \checkmark$ Roots are $2+i, 2-i, -3$. Factors are $(z+3)(z^2-4z+5)$ $= z^3 - z^2 - 7z + 15$ $\therefore a = -7, b = 15 \quad \checkmark \quad (4)$ c) $x^3 - 6x^2 + 12x - 35 = 0$. i) Roots $\alpha-2, \beta-2, \gamma-2$. Let $y = x-2$ i.e. $x = y+2$ $(y+2)^3 - 6(y+2)^2 + 12(y+2) - 35 = 0 \quad \checkmark$ $y^3 + 6y^2 + 12y + 8 - 6y^2 - 24y - 24 + 12y + 24 - 35 = 0$ $y^3 - 27 = 0 \quad \checkmark \quad (2)$	Other methods could be: Subst. $z = 2+i$ equate real & imag. parts, etc.

Suggested Solution (s)	Comments
<u>QUESTION 4 (cont.)</u> (c) (ii) $(y-3)(y^2+3y+9) = 0$ $y = 3 \text{ or } \frac{-3 \pm \sqrt{9-36}}{2} \quad \checkmark$ $y = 3 \text{ or } y = \frac{-3+3\sqrt{3}i}{2} \text{ or } y = \frac{-3-3\sqrt{3}i}{2}$ $x = 5, \frac{1}{2} + \frac{3\sqrt{3}}{2}i, \frac{1}{2} - \frac{3\sqrt{3}}{2}i \quad \checkmark \quad (2)$ (d) $z^2 + 5z - 2i = 0$. Roots $\alpha, \beta, \alpha+\beta$. $\Sigma \alpha: \alpha + \beta + (\alpha + \beta) = 2(\alpha + \beta)$ $= 2 \times (-5)$ $= -10 \quad \checkmark$ $\Sigma \alpha\beta: \alpha\beta + \alpha(\alpha+\beta) + \beta(\alpha+\beta)$ $= 3\alpha\beta + \alpha^2 + \beta^2$ $= (\alpha+\beta)^2 + \alpha\beta$ $= (-5)^2 + (-2i)$ $= 25 - 2i \quad \checkmark$ $\alpha\beta\gamma: \alpha\beta(\alpha+\beta) = (-2i)(-5)$ $= 10i \quad \checkmark$ $\therefore$ Cubic equation with roots $\alpha, \beta, \alpha+\beta$: $z^3 + 10z^2 + (25-2i)z - 10i = 0 \quad \checkmark \quad (4)$	OR 1- for $x=5$ by any method (eg. factor theorem) 1- other values correct $\rightarrow$ or equivalent $\rightarrow$ or equivalent. OR $\alpha + \beta = -5$ Equation is $(z+5)(z^2+5z-2i) = 0$ $z^3 + 5z^2 - 2iz + 5z^2 + 25z - 10i = 0$ i.e. $z^3 + 10z^2 + (25-2i)z - 10i = 0$

Suggested Solution (s)

Comments

QUESTION 5.

a) $\frac{x^2}{4} - \frac{y^2}{16} = 1$

$a^2 = 4$

$b^2 = 16$

(i) $b^2 = a^2(e^2 - 1)$

$a = 2$

$16 = 4(e^2 - 1)$

$b = 4$

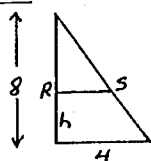
$e = \sqrt{5}$

(ii) Asymptotes: $y = \pm \frac{b}{a}x$

$y = \pm 2x$

b)(i) End view: $\frac{RS}{4} = \frac{8-h}{8}$

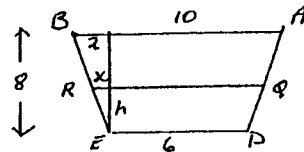
$RS = \frac{4}{8}(8-h)$
 $= 4 - \frac{1}{2}h$



Back surface:

$\frac{x}{2} = \frac{h}{8}$
 $x = \frac{h}{4}$

$RQ = 6 + 2 \times \frac{h}{4}$
 $= 6 + \frac{h}{2}$



$\therefore \text{Area PQRS} = (6 + \frac{h}{2})(4 - \frac{1}{2}h)$

(ii) $V = \int_0^8 (6 + \frac{h}{2})(4 - \frac{1}{2}h) dh$
 $= \int_0^8 (24 - h - \frac{h^2}{4}) dh$
 $= [24h - \frac{1}{2}h^2 - \frac{1}{12}h^3]_0^8$
 $= (24 \times 8 - \frac{1}{2} \times 64 - \frac{1}{12} \times 512)$

Volume = $117 \frac{1}{3} \text{ cm}^3$

Suggested Solution (s)

Comments

QUESTION 5 (cont.)

(c) $y = \frac{x^2 - 3x}{x+1}$

(i) $x+1 \overline{) x^2 - 3x + 0}$
 $\underline{x^2 + x}$
 $-4x + 0$
 $\underline{-4x - 4}$
 4

$y = x - 4 + \frac{4}{x+1}$

Asymptotes: $x = -1$, $y = x - 4$

(ii) $\frac{dy}{dx} = \frac{(x+1)(2x-3) - (x^2-3x)(1)}{(x+1)^2}$

$= \frac{2x^2 + 2x - 3}{(x+1)^2}$

$= \frac{(x+3)(x-1)}{(x+1)^2}$

stat. pts. at $x = -3$, $x = 1$

Stat. pts. at $(-3, -9)$, $(1, -1)$

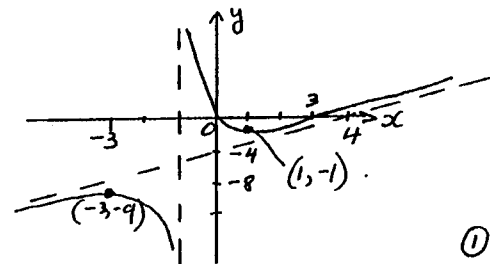
$f'(-3-\epsilon) = \frac{(-)(-)}{(+)} > 0$; $f'(-3+\epsilon) = \frac{(+)(-)}{(+)} < 0$

$\therefore$ Rel. maximum at $(-3, -9)$

$f'(1-\epsilon) = \frac{(+)(-)}{(+)} < 0$; $f'(1+\epsilon) = \frac{(+)(+)}{(+)} > 0$

$\therefore$ Rel. minimum at $(1, -1)$

(iii).



(iv). Two real roots for $k > -1$, $k < -9$

Suggested Solution (s)

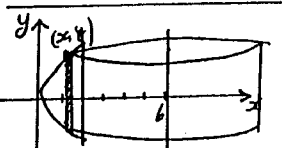
Comments

QUESTION 6

2) (i) $f(x) = x\sqrt{4-x^2}$
 $f(-x) = (-x)\sqrt{4-(-x)^2} = -x\sqrt{4-x^2} = -f(x)$
 $\therefore f(x)$ is odd function (1)

(ii) $\int_{-2}^2 x\sqrt{4-x^2} dx = 0$ ✓ since $f(x)$ is odd.
 $\int_{-2}^2 \sqrt{4-x^2} dx = \frac{1}{2} \times \pi \times 2^2 = 2\pi$ ✓
 since it represents area of a semi-circle

$\therefore \int_{-2}^2 (x\sqrt{4-x^2} - \sqrt{4-x^2}) dx = 0 - 2\pi = -2\pi$ (2)

b).  $y^2 = 4x$
 $y = \pm 2\sqrt{x}$

Radius = $6 - x$. ✓

$\delta V \doteq 2\pi(6-x)2y \delta x$

$\doteq 4\pi(6-x)y \delta x$ ✓

$V = 4\pi \int_0^2 (6-x)y dx$

$= 4\pi \int_0^2 (6-x)2\sqrt{x} dx$ ✓

$= 4\pi \int_0^2 (12x^{\frac{1}{2}} - 2x^{\frac{3}{2}}) dx$

$= 4\pi \left[12 \times \frac{2}{3} x^{\frac{3}{2}} - 2 \times \frac{2}{5} x^{\frac{5}{2}} \right]_0^2$ ✓

$= 4\pi \left[8 \times 2\sqrt{2} - \frac{4}{5} \times 2^{\frac{5}{2}} \right]$

$= 4\pi \left[80\sqrt{2} - 16\sqrt{2} \right]$

Volume = $\frac{256\sqrt{2}\pi}{5} \text{ unit}^3$ ✓

Suggested Solution (s)

Comments

QUESTION 6 (cont.)

(c) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(i) $\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$ ✓

At $P(a \cos \theta, b \sin \theta)$, $\frac{dy}{dx} = -\frac{b^2 a \cos \theta}{a^2 b \sin \theta}$
 $= -\frac{b \cos \theta}{a \sin \theta}$

Tangent: $y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$ ✓

$(a \sin \theta)y - ab \sin^2 \theta = -(b \cos \theta)x + ab \cos^2 \theta$

$(b \cos \theta)x + (a \sin \theta)y = ab(\cos^2 \theta + \sin^2 \theta)$

$(b \cos \theta)x + (a \sin \theta)y - ab = 0$. ✓

(3)

(ii) $S(ae, 0)$ $S'(-ae, 0)$

$SQ \times S'R = \left| \frac{(b \cos \theta)(ae) - ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \right| \times \left| \frac{(b \cos \theta)(-ae) - ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \right|$ ✓

$= \left| \frac{ab(e \cos \theta - 1)(-ab)(e \cos \theta + 1)}{b^2 \cos^2 \theta + a^2 \sin^2 \theta} \right|$

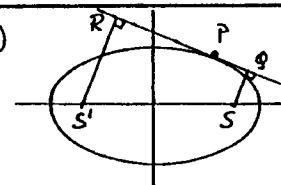
$= \left| \frac{a^2 b^2 (e^2 \cos^2 \theta - 1)}{a^2 (1 - e^2) \cos^2 \theta + a^2 \sin^2 \theta} \right|$ ✓

$= \left| \frac{b^2 (e^2 \cos^2 \theta - 1)}{\cos^2 \theta - e^2 \cos^2 \theta + \sin^2 \theta} \right|$ ✓

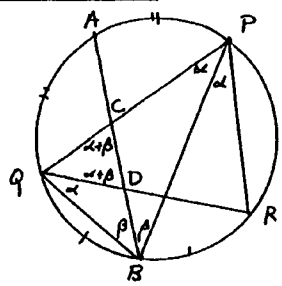
$= \left| \frac{b^2 (e^2 \cos^2 \theta - 1)}{1 - e^2 \cos^2 \theta} \right|$

$= b^2$ ✓

(4)



Suggested Solution (s)	Comments
<u>QUESTION 7.</u> a) $\cos \theta \geq \frac{1}{2}$ $2n\pi - \frac{\pi}{3} \leq \theta \leq 2n\pi + \frac{\pi}{3}$ (2) b) $y = mx$; $(x-a)^2 + (y-b)^2 = a^2 + b^2$ (i) $x^2 + y^2 - 2ax - 2by = 0$ Sub. $y = mx$: $x^2 + m^2x^2 - 2ax - 2bmx = 0$ $x^2(1+m^2) - x(2a+2bm) = 0$ ✓ $x = 0$ or $x = \frac{2a+2bm}{1+m^2}$ ✓ x-coord of P is $\frac{2(a+bm)}{1+m^2}$ (2) (ii) y-coord. of P is $\frac{2m(a+bm)}{1+m^2}$ ✓ $M: \left(\frac{a+bm}{1+m^2}, \frac{m(a+bm)}{1+m^2} \right)$ ✓ (2) (iii) $x = \frac{a+bm}{1+m^2}$ $y = mx$ $\therefore m = \frac{y}{x}$ $x(1+m^2) = a+bm$ $x\left(1 + \frac{y^2}{x^2}\right) = a + b\left(\frac{y}{x}\right)$ ✓ $x^2 + y^2 = ax + by$ ✓ $\left(x^2 - ax + \frac{a^2}{4}\right) + \left(y^2 - by + \frac{b^2}{4}\right) = \frac{a^2}{4} + \frac{b^2}{4}$ $\left(x - \frac{a}{2}\right)^2 + \left(y - \frac{b}{2}\right)^2 = \frac{a^2+b^2}{4}$ ✓ Locus is a circle, centre $\left(\frac{a}{2}, \frac{b}{2}\right)$ radius $\frac{\sqrt{a^2+b^2}}{2}$ ✓ (4)	1 mark for part of the solution eg $\frac{5\pi}{3} \leq \theta \leq \frac{7\pi}{3}$ or equivalent.

Suggested Solution (s)	Comments
<u>QUESTION 7 (cont.).</u> (c).  (i) $\angle QPB = \angle BPR$ because angles at the <u>circumference</u> standing on equal arcs are equal. (1) (ii) Let $\angle QPB = \angle BPR = \alpha$ ^{at circumference} $\angle BQR = \angle BPR = \alpha$ (angles on same arc) ✓ $\angle AQB = \angle ABP = \beta$ (angles at circumference on equal arcs) ✓ $\angle QDC = \angle QBD + \angle DQB$ (ext. angle $\triangle QDB$) ✓ $= \beta + \alpha$ $\angle QCD = \angle CPB + \angle CBP$ (ext. angle $\triangle BCP$) ✓ $= \alpha + \beta$ $\therefore \angle QDC = \angle QCD$ $\therefore QC = QD$. (4)	Must mention "circumference"

Suggested Solution (s)

Comments

QUESTION 8

$$a) x = V \cos \theta t, y = V \sin \theta t - \frac{1}{2} g t^2$$

(i) Subst. $x = d$, $y = h$ and eliminate t .

$$d = V \cos \theta t, h = V \sin \theta t - \frac{1}{2} g t^2$$

$$t = \frac{d}{V \cos \theta}, h = V \sin \theta \cdot \frac{d}{V \cos \theta} - \frac{g}{2} \times \frac{d^2}{V^2 \cos^2 \theta}$$

$$h = d \tan \theta - \frac{g d^2}{2 V^2 \cos^2 \theta} \quad \checkmark$$

$$\frac{g d^2}{2 V^2 \cos^2 \theta} = d \tan \theta - h$$

$$V^2 = \frac{g d^2}{2 \cos^2 \theta (d \tan \theta - h)} \\ = \frac{g d^2 \sec^2 \theta}{2 (d \tan \theta - h)} \quad (2)$$

$$(ii) y = V \sin \theta - g t \\ = 0 \text{ when } t = \frac{V \sin \theta}{g} \quad \checkmark$$

$$\text{Max. ht.} = V \sin \theta \times \frac{V \sin \theta}{g} - \frac{g}{2} \times \frac{V^2 \sin^2 \theta}{g^2}$$

$$= \frac{V^2 \sin^2 \theta}{2g} \quad \checkmark$$

$$= \frac{g d^2 \sec^2 \theta}{2 (d \tan \theta - h)} \times \frac{\sin^2 \theta}{2g} \quad (3)$$

$$= \frac{d^2 \tan^2 \theta}{4 (d \tan \theta - h)} \quad \checkmark \text{ since } \sec^2 \theta \sin^2 \theta \\ = \frac{1}{\cos^2 \theta} \sin^2 \theta = \tan^2 \theta$$

$$(iii) h = \frac{d^2 \tan^2 \theta}{4 (d \tan \theta - h)} \quad \checkmark$$

$$d^2 \tan^2 \theta = 4h (d \tan \theta - h)$$

$$d^2 \tan^2 \theta - 4dh \tan \theta + 4h^2 = 0$$

$$(d \tan \theta - 2h)^2 = 0 \quad \checkmark$$

$$d \tan \theta - 2h = 0$$

$$\therefore \tan \theta = \frac{2h}{d} \quad \checkmark \quad (3)$$

OR Subst. $\tan \theta = \frac{2h}{d}$
into (iii) and
show it equals h .
Need description
conclusion for
third mark.

Suggested Solution (s)

Comments

QUESTION 8 (cont.)

(b)(i) Prove $(\sqrt{3}-1)^n = p_n + q_n \sqrt{3}$; p_n, q_n integers

$$\text{When } n=1, \text{ LHS} = (\sqrt{3}-1)^1 = -1 + \sqrt{3} \quad \checkmark$$

$p_1 = -1, q_1 = 1$ are unique $\therefore$ true for $n=1$

Assume it is true for $n=k$,

i.e. assume $(\sqrt{3}-1)^k = p_k + q_k \sqrt{3}$ (p_k, q_k unique integers) $\checkmark$

When $n=k+1$,

$$\text{LHS} = (\sqrt{3}-1)^{k+1}$$

$$= (\sqrt{3}-1)(\sqrt{3}-1)^k$$

$$= (\sqrt{3}-1)(p_k + q_k \sqrt{3}) \quad \checkmark \text{ by assumption}$$

$$= p_k \sqrt{3} + 3q_k - p_k - q_k \sqrt{3}$$

$$= (3q_k - p_k) + (p_k - q_k) \sqrt{3}$$

$$= p_{k+1} + q_{k+1} \sqrt{3} \quad \checkmark$$

$$\text{where } p_{k+1} = 3q_k - p_k, q_{k+1} = p_k - q_k$$

$\therefore$ If it is true for $n=k$, it is true for $n=k+1$

Since it is true for $n=1$, it is true for $n=2, 3, \dots$ $\checkmark$

Note: Mark for
conclusion not
awarded if second
last mark not
awarded

$$(ii) p_n^2 - 3q_n^2 = (3q_{n-1} - p_{n-1})^2 - 3(p_{n-1} - q_{n-1})^2$$

$$= 9q_{n-1}^2 - 6q_{n-1}p_{n-1} + p_{n-1}^2$$

$$- 3p_{n-1}^2 + 6p_{n-1}q_{n-1} - 3q_{n-1}^2$$

$$= 6q_{n-1}^2 - 2p_{n-1}^2$$

$$= -2(p_{n-1}^2 - 3q_{n-1}^2) \quad \checkmark$$

$$= (-2)(-2)(p_{n-2}^2 - 3q_{n-2}^2)$$

$$= (-2)(-2) \dots (-2)(p_1^2 - 3q_1^2)$$

$$= (-2)(-2) \dots (-2)(1-3)$$

$$= (-2)^n \quad \checkmark \quad (2)$$

OR
Prove by induction
(but only 2 marks).